Notes: Reher & Valkanov (JF, 2024)

The Mortgage-Cash Premium Puzzle

Contents

1 Big picture

- **Question.** Why do mortgage-financed homebuyers pay more than all-cash buyers for otherwise comparable homes?
- **Object.** The paper studies the *mortgage-cash premium*: the expected log price difference between a mortgage-financed offer and an all-cash offer for the same home.
- **Main empirical fact.** All-cash purchases are about one-third of U.S. home purchases from 1980 to 2017, and all-cash purchase prices are much lower than mortgage-financed purchase prices in raw data.
- **Main estimate.** Using repeat-sales and hedonic controls, the authors estimate an average mortgage-cash premium of about 11% to 12%.
- **Why this is puzzling.** A representative-seller model calibrated to average transaction frictions implies a premium of only about 3.3%.
- **Core friction.** Mortgaged offers expose sellers to mortgage-origination frictions: approval risk, appraisal risk, delay, possible transaction failure, and relisting costs.
- **Selection challenge.** Mortgage financing is not randomly assigned. Mortgaged and all-cash buyers may differ in wealth, bargaining skill, property targets, seller motivation, and property condition.
- **Robustness strategy.** The paper uses multiple research designs: repeat-sales hedonic regressions, offer-level data, seller/buyer controls, IV designs, listing controls, semistructural estimators, matching, and external data from CoreLogic.
- **Heterogeneity result.** Accounting for heterogeneity in selling conditions raises the theoretical premium to about 6.9%, explaining roughly half of the apparent puzzle.
- **Where the puzzle remains.** The gap between empirical and theoretical premiums is largest when transaction-failure risk is moderate or high.
- **Survey result.** A survey experiment of homeowners replicates the pattern and suggests that belief distortions or ambiguity about failure risk can amplify the premium in high-risk states.
- **Economic takeaway.** Mortgage transaction frictions and seller beliefs create a large price wedge between mortgage-financed and cash-financed buyers. The wedge is not merely a selection artifact.

2 Data and measurement

2.1 Baseline transaction data

- The baseline data set is Zillow ZTRAX, covering U.S. home purchases from 1980 to 2017.
- The key dependent variable is the log real sales price of a home purchase.
- The key treatment variable is $Mortgaged_{i,t}$, an indicator equal to one when the loan amount is positive.
- The baseline repeat-sales sample contains 426,256 transactions.
- The broad transaction-level data contain millions of purchases, and the repeat-sales sample enables property fixed effects.

Interpretation: The repeat-sales structure is important because it compares the same property at different times, reducing bias from time-invariant property quality.

2.2 All-cash and mortgaged purchases

- All-cash purchases are identified as purchases with no positive loan amount.
- Mortgaged purchases are identified as purchases with a positive loan amount.
- The paper documents a large mass of cash-financed purchases and a large raw price gap between cash and mortgaged purchases.
- The raw price gap is not interpreted causally because cash buyers may buy cheaper or different homes.

Interpretation: The raw facts motivate the mortgage-cash premium but do not identify it. Identification requires comparing comparable transactions.

2.3 Offer-level data

- The authors use offer-level data from Redfin for January 2020 through June 2021.
- Each observation is an offer to purchase a home.
- Variables include whether the offer is mortgaged, whether it wins, the offer price, and the total number of offers.
- The baseline offer-level sample contains 22,516 offers.

Interpretation: Offer-level data allow the authors to compare accepted and nonaccepted offers on the same home, directly addressing buyer selection and bargaining concerns.

2.4 Seller and buyer controls

- Buyer controls include institutional buyer status, foreign buyer status, same-county buyer status, and cash-purchase propensity.
- Seller controls include seller loan-to-value, same-month seller purchase, foreign seller status, cash-sale share, and number of previous sales.
- Property controls include age, rooms, bathrooms, stories, air conditioning, detached status, and flip indicators.

Interpretation: These variables proxy for buyer wealth, seller urgency, property condition, and bargaining or screening motives.

2.5 Experimental survey data

- The authors conduct a homeowner survey experiment with multiple waves.
- Respondents face hypothetical choice tasks between all-cash and mortgage-financed offers.
- The experiment varies transaction-failure probability and whether the failure rate is ambiguous or known.
- The survey elicits both the experimental mortgage-cash premium and subjective beliefs about failure risk.

Interpretation: The survey isolates seller beliefs and nonstandard preferences more directly than observational transaction data.

3 Models and empirical specifications

3.1 Back-of-envelope benchmark

The simplest risk-neutral logic equates the expected benefit of a mortgage premium to the expected cost of failure:

$$0 = (1 - q) \times Premium - q \times \kappa, \tag{1}$$

where q is the failure probability and κ is the cost of failure.

Interpretation: With a 6% failure probability and a 10% failure cost, the implied premium is only about 0.6%. This illustrates why the empirical 11% premium is surprising.

3.2 Representative-seller model

The full model embeds:

- risk-averse sellers;
- mortgage transaction failure and delay;
- seller financial wealth and down payment needs;
- outstanding seller mortgage balances;
- offer arrival rates and bargaining competition;
- home maintenance costs and relisting costs;
- possible nonfinancial contingencies.

Interpretation: The calibrated representative-seller model produces a 3.3% premium, substantially below the empirical estimate.

3.3 Baseline repeat-sales hedonic regression

The main empirical equation is

$$\log(Price_{i,t}) = \mu Mortgaged_{i,t} + \psi X_{i,t} + \zeta_{z(i),t} + \alpha_i + \epsilon_{i,t}. \tag{2}$$

- μ is the mortgage-cash premium.
- $\zeta_{z(i),t}$ are zip-code-by-month fixed effects.
- α_i are property fixed effects.
- $X_{i,t}$ contains hedonic and transaction controls.

Interpretation: The coefficient μ compares the same property across transactions, within local time conditions, under different methods of financing.

3.4 Offer-level regression

The offer-level design estimates

$$\log(Price_{i,j,t}) = \mu\,(Mortgaged_{i,j,t} \times Winning_{i,j,t}) + \psi_0 Winning_{i,j,t} + \psi_1 Mortgaged_{i,j,t} + \zeta_{z(i)} + \tau_t + u_{i,j,t}. \tag{3}$$

Interpretation: Comparing winning and losing offers on the same home helps address buyer valuation, bargaining power, and property-targeting concerns.

3.5 Heterogeneity regression

The paper estimates premiums across terciles of model-implied selling conditions:

$$\log(Price_{i,t}) = \mu_0 Mortgaged_{i,t} + \sum_{\theta,p} \mu_{\theta,p} \left(Mortgaged_{i,t} \times Tercile(\widehat{\theta}_{z,t}, p) \right) + \psi X_{i,t} + \zeta_{z(i),t} + \alpha_i + \epsilon_{i,t}. \quad (4)$$

Interpretation: This connects empirical premium heterogeneity to theoretical model parameters such as failure risk, seller wealth, offer arrival rates, and buyer leverage capacity.

3.6 Survey heterogeneity regression

The experimental analogue relates respondent premiums to elicited or assigned selling conditions:

$$Premium_k = \sum_p \mu_{q,p} Tercile(\widehat{q}_k, p) + \sum_p \Delta\mu^A_{q,p} Tercile(\widehat{q}_k, p) Ambiguity_k + \sum_{\theta,p} \mu_{\theta,p} Tercile(\widehat{\theta}_k, p) + \epsilon_k. \quad (5)$$

Interpretation: The ambiguity interaction tests whether sellers demand higher premiums when failure probabilities are uncertain rather than known.

4 Identification and empirical design

4.1 Main identification problem

- Mortgaged buyers and all-cash buyers may buy different homes.
- Cash buyers may target distressed sellers or low-quality properties.
- Mortgaged buyers may have higher willingness-to-pay or different preferences.
- Sellers may accept lower cash offers because cash buyers are faster, safer, or better at negotiation.

Interpretation: The central identification problem is separating a true seller compensation premium from selection into financing type.

4.2 Why property and local-time fixed effects matter

- Property fixed effects absorb time-invariant home quality.
- Zip-code-by-month fixed effects absorb local housing-market conditions.
- Hedonic controls absorb observable changes in home characteristics.
- Repeat-sales design reduces bias from persistent unobserved property traits.

Interpretation: The baseline design identifies the premium from within-property changes in financing status under the same local market conditions.

4.3 Systematic selection-bias checks

- **Buyer selection:** offer-level data compare accepted and nonaccepted offers.
- **Seller selection:** seller controls, listing data, and IV designs account for seller urgency and willingness to accept cash.
- **Property selection:** new homes, no-flip samples, semistructural estimators, and matching address property-condition concerns.

- **External validity:** nonrepeat sales, CoreLogic data, and inverse-probability weighting assess representativeness.

Interpretation: The estimate remains between roughly 8% and 17% across many designs, with the baseline around 11%.

4.4 What remains imperfect

- The baseline data do not directly observe every offer considered by every seller.
- Some property condition changes may remain unobserved.
- Seller beliefs and ambiguity aversion are not directly observed in transaction data.
- The theoretical model is rich but still stylized.

Interpretation: The paper's strength is triangulation: many imperfect designs point to the same large premium.

5 Tables and figures

5.1 Figure 1 + Table I: cash-purchase facts and representative-seller calibration

Read:

- Figure 1 shows a large mass of cash transactions and bunching at high loan-to-value ratios among mortgaged transactions.
- Cash purchases have much lower average log prices than mortgaged purchases across the full sample and subsamples.
- Table I reports the representative-seller model calibration.
- The representative-seller premium is 3.3%, while the seller cost of failure is 4.7% and the failure probability is 6.4%.

Economic message: The raw facts suggest sellers value cash financing. The representative model implies that average frictions alone are too small to match the empirical premium.

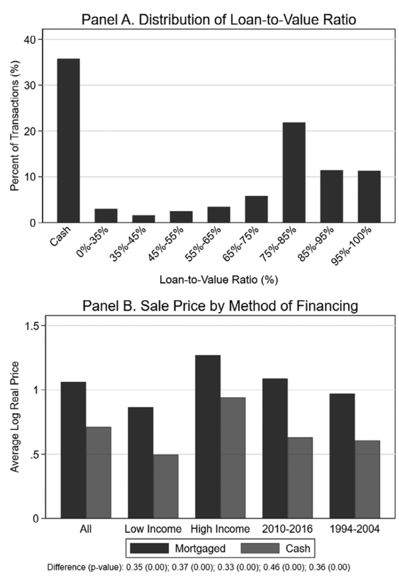

Figure 1. Facts about all-cash purchases. This figure documents two facts about cash-financed home purchases. Panel A plots the distribution of LTV ratios across purchases over 1980 to 2017. Panel B plots the average log sales price in hundreds of thousands of 2010 dollars for purchases financed by a mortgage (Mortgaged) and exclusively with cash (Cash). The figure plots this average for different subsamples: All denotes all observed purchases, Low Income and High Income denote purchases in zip codes in the lowest and highest quartile by 2010 income, respectively, and 2010 to 2016 and 1994 to 2004 denote purchases over these two periods. Data are from ZTRAX.

(a) Figure 1: cash purchase facts.

Table I

Representative Seller Calibration of the Mortgage-Cash Premium

This table summarizes the notation of the baseline model in Section I and performs a stylized calibration of the mortgage-cash premium for a representative home seller. The endogenous variables are the mortgage-cash premium, μ, and the seller's endogenous cost of failure, κ, implied by this stylized calibration. The expressions for μ and κ are shown in Proposition 1 as a function of parameters $\Theta = \{\Theta_1; \Theta_2\}$. The elements in Θ_1 are exogenous and are used to calculate the values of the endogenous variables. These values equal the averages across the distribution of the empirical measures obtained from a more rigorous calibration, summarized in Table VII. The elements in Θ_2 are also exogenous and are externally calibrated. Under a low-friction parameterization in which $\xi = \ell_s = q^d = m = 0$, $\lambda = 20$, $\tilde{\delta} = 0.01$, and the other parameters are as in the table, we obtain $\mu = 0.009$ and $\kappa = 0.009$, as referenced in Section I.E. The model's setup and solution are in Section I.

Parameter	Notation	Value	Reference
Endogenous:			
Mortgage-cash premium	μ	0.033	Equation (19)
Seller cost of failure	κ	0.047	Equation (20)
Exogenous (Θ_1):			
Probability of transaction failure	q	0.064	HMDA Denial
Offer arrival rate	λ	2.162	Offer-Level Data
Seller financial wealth-to-housing	w	1.577	SCF Total Wealth
Seller down payment	ξ	0.052	Zillow, ZTRAX
Seller loan-to-value ratio	ℓ_s	0.294	Zillow, ZTRAX
Buyer liquid assets-to-housing	y	0.162	IRS Income
Loan-to-value constraint	$\bar{\ell}$	0.864	Zillow, ZTRAX
Probability of transaction delay	q^d	0.30	NAR Index
Mortgaged offer share	m	0.622	Offer-Level Data
Maintenance cost-to-value, annualized	$\tilde{\delta}$	0.056	Zillow, ZORI
Preferences (Θ_2):			
Coefficient of relative risk aversion	γ	5	
Discount rate, annualized	ρ	0.04	
Borrower loan cost, share of par	D	1	

(b) Table I: representative-seller calibration.

5.2 Table II + Table III: data summary and financing choice

Read:

- Table II summarizes the ZTRAX transaction-level and Redfin offer-level data.
- In transaction data, 64.2% of purchases are mortgaged and the mean real price is about $416,813.
- In offer data, 60.9% of offers are mortgaged and the mean offer price is about $512,662.
- Table III shows that high denial rates reduce mortgage financing, while high transaction volume and high average house prices increase it across markets.

- Within markets, institutional buyers, foreign buyers, and flips are less likely to be mortgaged.

Economic message: The financing method is strongly related to buyer, seller, property, and market characteristics, which motivates the paper's rich identification strategy.

Table II

Summary of Variables in the Empirical Analysis

This table summarizes variables from our observational data sets. Panel A summarizes variables from our core data set, the ZTRAX data set. Subscripts i and t index property and month. Each observation is a home purchase transaction over 1980 to 2017. The variables are defined as follows: $RealPrice_{i,t}$ is the sales price in 2010 dollars, $Mortgaged_{i,t}$ indicates whether the loan amount is positive, Age_i is the number of years from when the property was built, $Rooms_i$ through $Stories_i$ are the number of overall rooms, bathrooms, and stories, respectively, $AirConditioning_i$ and $Detached_i$ indicate if i has air conditioning and is a detached single-family home, respectively, $Flip_{i,t}$ indicates whether the property is subsequently sold within 12 months, $ForeignBuyer_{i,t}$ indicates whether the buyer has a foreign address, $Same-CountyBuyer_{i,t}$ indicates whether the buyer's address is in the same county as the property, $InstitutionalBuyer_{i,t}$ indicates whether the buyer is an institution and the property is not to be owner occupied, $CashPropensity_{b(i,t)}$ indicates whether the buyer of property i in month t, denoted $b(i,t)$, buys another home all-cash over our sample period, $HighSellerLTV_{i,t}$ indicates whether the seller's LTV ratio is above 50%, where the numerator is imputed using a straight-line amortization according to loan term and the denominator is imputed using the median sales price in the buyer's zip code, $Same-MonthSellerPurchase_{i,t}$ indicates whether the seller purchases another home in the same month, $ForeignSeller_{i,t}$ indicates whether the seller has a foreign address, $CashShare_{s(i,t)}$ is the share of homes sold to cash buyers over our sample period by the seller of property i in month t, denoted $s(i,t)$, after excluding the sale in question and assigning a value of zero to sellers who appear only once in the data, and $NumberofSales_{s(i,t)}$ equals the number of sales made by the seller as of month t in the baseline ZTRAX data set. With the exception of $Mortgaged_{i,t}$, all indicator variables are assigned a value of zero when the raw variable is unobserved. Panel B summarizes variables from the offer-level data set. Each observation is an offer to purchase a home over the period from January 2020 through June 2021 made through a real estate agent affiliated with Redfin. The variables are defined as follows: $Mortgaged_{i,j,t}$ indicates whether j is an offer with a positive loan amount, $Winning_{i,j,t}$ indicates whether j is the winning offer, $RealPrice_{i,j,t}$ is the price offered by j in 2010 dollars, and $NumberofOffers_{i,j,t}$ is the total number of offers, including j. The bottom rows show the number of observations from each data set that can be used in Tables IV and VI, respectively. Section II and Internet Appendix Section I contain additional details.

Variable	Mean	Standard Deviation	Variable	Mean	Standard Deviation
Panel A: Transaction-Level Data Set					
$Real\ Price_{i,t}$	\$416,813	\$776,315	$Mortgaged_{i,t}$	0.642	0.479
Age_i	28.431	6.682	$Rooms_i$	1.409	1.492
$Bathrooms_i$	0.237	0.715	$Stories_i$	1.096	0.111
$Air\ Conditioning_i$	0.217	0.239	$Detached_i$	0.405	0.491
$Flip_{i,t}$	0.115	0.319	$Foreign\ Buyer_{i,t}$	0.003	0.054
$Same\text{-}County\ Buyer_{i,t}$	0.009	0.093	$Institutional\ Buyer_{i,t}$	0.001	0.038
$Cash\ Propensity_{b(i,t)}$	0.252	0.434	$High\ Seller\ LTV_{i,t}$	0.068	0.252
$Same\text{-}Month\ Seller\ Purchase_{i,t}$	0.015	0.122	$Foreign\ Seller_{i,t}$	0.002	0.041
$Cash\ Share_{s(i,t)}$	0.179	0.327	$Number\ of\ Sales_{s(i,t)}$	4.687	1.034
Panel B: Offer-Level Data Set					
$Real\ Price_{i,j,t}$	\$512,662	\$359,480	$Mortgaged_{i,j,t}$	0.609	0.488
$Winning_{i,j,t}$	0.166	0.372	$Number\ of\ Offers_{i,j,t}$	5.787	5.120

Baseline Number of Transactions: 426,256
Baseline Number of Offers: 22,516

(a) Table II: summary variables.

Table III

Characteristics of Mortgaged Transactions

This table correlates the probability that a transaction is mortgage financed with market-level and transaction-level characteristics. P-values are in parentheses. Subscripts i, z, and t index property, zip code, and month. $Mortgaged_{i,t}$ indicates whether the loan amount is positive. Transaction-level variables are defined in Table II. Market-level variables are observed at the zip code-by-year level: $MortgageDenialRate_{z,t}$ denotes the mortgage application denial rate, using data from HMDA, $TransactionVolume_{z,t}$ is the total number of home purchase transactions in the zip code and year in the ZTRAX data set, $AverageHousePrice_{z,t}$ is the FHFA All-Transactions house price index, and $AverageIncome_{z,t}$ comes from the IRS data set. Columns (3) and (4) include a zip code-by-month fixed effect and so exclude market-level variables. Transaction-level variables are included in columns (1) and (2) but not tabulated. The sample period is 1980 to 2017. Standard errors are clustered by property.

Outcome:	$Mortgaged_{i,t}$			
	Across Markets		Within Markets	
	(1)	(2)	(3)	(4)
Market level:				
$Mortgage\ Denial\ Rate_{z,t}$	−0.318	−0.156		
	(0.018)	(0.017)		
$\log(Transaction\ Volume_{z,t})$	0.055	0.045		
	(0.001)	(0.003)		
$\log(Average\ House\ Price_{z,t})$	0.097	0.046		
	(0.002)	(0.004)		
$\log(Average\ Income_{z,t})$	0.003	0.008		
	(0.002)	(0.004)		
Transaction level:				
$Institutional\ Buyer_{i,t}$			−0.494	−0.656
			(0.002)	(0.003)
$Foreign\ Buyer_{i,t}$			−0.125	−0.148
			(0.021)	(0.044)
$Flip_{i,t}$			−0.080	−0.086
			(0.002)	(0.002)
$High\ Seller\ LTV_{i,t}$			0.031	−0.134
			(0.002)	(0.002)
$Foreign\ Seller_{i,t}$			−0.050	−0.138
			(0.031)	(0.066)
$Institutional\ Seller_{i,t}$			0.057	0.037
			(0.002)	(0.003)
Month FE	Yes	Yes	No	No
Zip code FE	No	Yes	No	No
Zip code-month FE	No	No	Yes	Yes
Property FE	No	No	No	Yes
R^2	0.300	0.461	0.628	0.873
Number of observations	425,389	425,389	425,386	425,370

(b) Table III: characteristics of mortgaged transactions.

5.3 Table IV + Figure 2: baseline empirical premium and time-series pattern

Read:

- Table IV estimates the baseline repeat-sales hedonic regression.
- The baseline repeat-sales estimate is 11.7%.
- Estimates by period remain high: 13.1% in 1980–1996, 12.4% in 1996–2004, 16.5% in 2005–2010, and 11.9% in 2010–2017.
- New homes, no-flip samples, noninstitutional samples, nonforeign samples, and positive-equity samples all yield positive premiums.
- Figure 2 shows that the premium is correlated over time with transaction-failure risk and the all-cash purchase share.

Economic message: The premium is large, persistent, and robust across major sample restrictions.

Table IV

Empirical Mortgage-Cash Premium

This table estimates equation (28), which calculates the price premium paid by mortgaged buyers relative to all-cash buyers (i.e., the mortgage-cash premium). P-values are in parentheses. Subscripts i and t index property and month. The regression equation is of the form

$$\log\left(Price_{i,t}\right) = \mu Mortgaged_{i,t} + \psi X_{i,t} + \zeta_{z(i),t} + \alpha_i + \epsilon_{i,t},$$

where observations are home purchases, $Mortgaged_{i,t}$ indicates whether the loan amount is positive, $Price_{i,t}$ is the sales price, α_i is a property fixed effect, $\zeta_{z(i),t}$ is a zip code-by-month fixed effect, and $X_{i,t}$ is a vector of indicators for whether i belongs to bins defined by month and the values of the following hedonic characteristics: the number of years from when the property was built, the number of overall rooms, bathrooms, and stories, an indicator for whether i has air conditioning, and an indicator for whether i is a detached single-family home. The sample period in columns (0) and (1) is 1980 to 2017. Columns (2) to (5) restrict the sample to months within the indicated time periods. Column (6) restricts the sample to properties built within the previous three years (New Homes). Column (7) restricts the sample to properties that were sold at least 12 months later (No Flips). Column (8) restricts the sample to properties in which neither the buyer nor seller is an institution (Non Instit.). Column (9) restricts the sample to properties in which neither the buyer nor seller has an address outside the United States (Non Foreign). Column (10) drops purchases in which the seller's LTV ratio is above 100% (Positive Equity), where the numerator is imputed using a straight-line amortization according to loan term and the denominator is imputed using the average sales price in the surrounding zip code and month to avoid mechanical correlation with $\log\left(Price_{i,t}\right)$. Standard errors are clustered by property. Data are from the ZTRAX data set.

Outcome:	$\log\left(Price_{i,t}\right)$										
	(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
$Mortgaged_{i,t}$	0.161 (0.000)	0.117 (0.000)	0.131 (0.000)	0.124 (0.000)	0.165 (0.000)	0.119 (0.000)	0.090 (0.000)	0.088 (0.000)	0.112 (0.000)	0.108 (0.000)	0.099 (0.000)
Sample restriction	None	None	1980–1996	1996–2004	2005–2010	2010–2017	New homes	No flips	Non instit.	Non foreign	Positive equity
Zip code-month FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Hedonic-month FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Property FE	No	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes
R^2	0.582	0.907	0.969	0.842	0.797	0.826	0.647	0.963	0.895	0.904	0.923
Number of observations	2,254,389	426,256	27,087	122,683	69,307	20,792	6,651	186,570	333,288	323,956	313,370

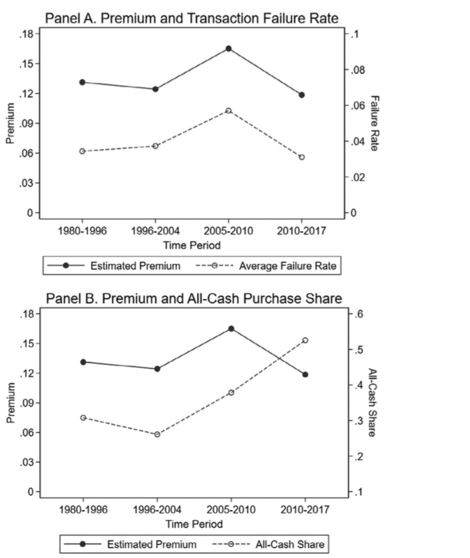

Figure 2. Empirical mortgage-cash premium over time. This figure plots the empirical mortgage cash premium over various time periods. The empirical premium is the value of μ that comes from estimating equation (28) on the subsample $\mathcal{T}$ consisting of months within the indicated time period,

$$\log\left(Price_{i,t}\right) = \mu Mortgaged_{i,t} + \psi X_{i,t} + \zeta_{z(i),t} + \alpha_i + \epsilon_{i,t}, \quad t \in \mathcal{T}⇔$$

(a) Figure 2: empirical premium over time.

Table V

Summary of Estimates of the Mortgage-Cash Premium

This table summarizes various estimates of the mortgage-cash premium. Details on each methodology are provided in the notes to the indicated table.

	Empirical Mortgage-Cash Premium ($\hat{\mu}$)	
	Estimate	Table
Repeat-sales-hedonic, baseline	0.117	Table IV, column (1)
Repeat-sales-hedonic, 1980-1996	0.131	Table IV, column (2)
Repeat-sales-hedonic, 1996 to 2004	0.124	Table IV, column (3)
Repeat-sales-hedonic, 2005 to 2010	0.165	Table IV, column (4)
Repeat-sales-hedonic, 2010 to 2017	0.119	Table IV, column (5)
Repeat-sales-hedonic, new homes	0.090	Table IV, column (6)
Repeat-sales-hedonic, no flips	0.088	Table IV, column (7)
Repeat-sales-hedonic, non institutional	0.112	Table IV, column (8)
Repeat-sales-hedonic, non foreign	0.108	Table IV, column (9)
Repeat-sales-hedonic, positive equity	0.099	Table IV, column (10)
Nonaccepted offers	0.086	Table VI, column (2)
Homeowner Survey, Wave 1	0.106	Table VIII, column (2)
Homeowner Survey, Wave 2	0.104	Table VIII, column (3)
Homeowner Survey, Wave 3	0.107	Table VIII, column (4)
Hedonic with nonrepeat sales	0.161	Internet Appendix Table IA.I, column (5)
Buyer and seller characteristics	0.129	Internet Appendix Table IA.II, column (3)
IV, regulatory appraisal floor	0.136	Internet Appendix Table IA.III, column (2)
IV, propensity to sell all-cash	0.139	Internet Appendix Table IA.IV, column (4)
CoreLogic Data Set	0.122	Internet Appendix Table IA.I, column (8)
Listing characteristics	0.143	Internet Appendix Table IA.VII, column (2)
Semistructural (Bajari et al. (2012))	0.149	Internet Appendix Table IA.VIII, column (2)
Weighting by representativeness	0.103	Internet Appendix Table IA.IX, column (2)
Matching	0.169	Internet Appendix Table IA.XI, column (3)

(b) Table V: summary of estimates.

5.4 Table V + Table VI: robustness and offer-level evidence

Read:

- Table V summarizes many estimates of the mortgage-cash premium.
- Estimates range from 8.6% using nonaccepted offers to 16.9% using matching.
- CoreLogic data produce a similar estimate of 12.2%.
- Table VI uses offer-level Redfin data and estimates an 8.1% to 8.6% premium.
- The offer-level estimate is a credible lower bound because it controls for winning offers and nonaccepted offers on the same property.

Economic message: Robustness checks consistently support a large premium, and offer-level data directly address buyer selection and bargaining concerns.

Table VI

Robustness to Using Data on Nonaccepted Offers

This table estimates equation (35), which assesses whether the baseline results are robust to using data on nonaccepted offers to estimate the mortgage-cash premium. *P*-values are in parentheses. Subscripts i, j, and t index property, offer, and month. The regression equation is of the form

$$\log(Price_{i,j,t}) = \mu\,(Mortgaged_{i,j,t} \times Winning_{i,j,t}) + \psi_0 Winning_{i,j,t} + \psi_1 Mortgaged_{i,j,t} + \zeta_{z(i)} + \tau_t + \upsilon_{i,j,t},$$

where observations are home purchase offers, $Mortgaged_{i,j,t}$ indicates whether j is an offer with a mortgage, $Winning_{i,j,t}$ indicates whether j is the offer that is accepted, $Price_{i,j,t}$ is the price offered by j, and $\zeta_{z(i)}$ and τ_t are zip code and month fixed effects, respectively. Column (2) includes a vector of fixed effects for the number of offers on the property. The sample consists of purchase offers made through Redfin real estate agents over January 2020 through June 2021. Observations are weighted by the number offers on the property. Standard errors are heteroskedasticity robust. Data are from the offer-level data set. The remaining notes are the same as in Table IV.

Outcome:	$\log(Price_{i,j,t})$	
	(1)	(2)
$Mortgaged_{i,j,t} \times Winning_{i,j,t}$	0.081	0.086
	(0.008)	(0.004)
Other variables:		
$Mortgaged_{i,j,t}$	−0.006	−0.006
	(0.304)	(0.279)
$Winning_{i,j,t}$	−0.044	−0.060
	(0.098)	(0.025)
Month FE	Yes	Yes
Zip FE	Yes	Yes
Offers-on-property FE	No	Yes
R^2	0.619	0.620
Number of observations	22,516	22,516

5.5 Table VII + Figure 3: distribution of parameters and theoretical premium

Read:

- Table VII reports the distribution of model parameters across markets and sellers.
- Failure probability, offer arrival rate, seller net wealth, down payment, LTV, buyer liquidity, and delay risk vary substantially.
- Figure 3 shows that averaging the model across the parameter distribution yields a 6.9% theoretical premium.
- Risk neutrality implies only 1.9%, and the representative-agent calibration yields 3.3%.
- Adding nonfinancial contingencies raises the theoretical premium to about 7.9%, while high risk aversion raises it to about 8.1%.

Economic message: Heterogeneity in selling conditions explains about half of the gap between the representative-seller model and the empirical premium.

5.6 Figure 4 + Figure 5: heterogeneity and extensions

Read:

- Figure 4 compares empirical and theoretical premiums across success-rate and seller-net-wealth terciles.
- Both empirical and theoretical premiums are high when the probability of transaction success is low.
- The empirical premium exceeds the theoretical premium especially in high-friction states.
- Figure 5 shows that nonfinancial contingencies and high risk aversion reduce but do not eliminate the gap.

Economic message: The puzzle is not uniform. It remains primarily in states where transaction risk is salient.

5.7 Table VIII + Figure 6: survey experiment and belief distortions

Read:

- Table VIII summarizes the homeowner survey experiment.

Table VII

Distribution of Model Parameters

This table reports the distributions of the empirical measures of the model's parameters. Panel A reports the data source for directly calibrated parameters and the midpoint of each tercile. Probability of Failure (q) is measured as the mortgage application denial rate among preapproved, first-lien, mortgages for the purchase of an owner-occupied, single-family home, based on the HMDA data set aggregated to the zip code-by-year level. Offer Arrival Rate (λ) is measured using the relationship $\mathbb{E}[N|N>0]=\frac{\lambda}{1-e^{-\lambda}}$. The number of offers received conditional on receiving an offer, $\mathbb{E}[N|N>0]$, is measured using the Offer-Level data set aggregated to the zip code level. Seller Wealth (w) is measured as the ratio of: the sum of checking and savings accounts, certificates of deposit, cash, stocks, savings bonds, other bonds, mutual funds, annuities, trusts, IRAs, employer-provided retirement plans, and total household income, divided by the value of the home. The unit of analysis is a household in the SCF data set, excluding nonhomeowners and first-time homeowners. Seller LTV (ℓ_s) is measured as the current LTV ratio, imputed from the ratio of mortgage balance implied by a straight-line amortization to the sale price. Seller Down Payment (ξ) is measured as the ratio of next-home price minus next-home mortgage to current-home price for sellers who purchase another home within 12 months. LTV Constraint ($\bar{\ell}$) is measured as the LTV ratio conditional on lying between the GSE regulatory limit and 100%. The calibration of ℓ_s, ξ, and $\bar{\ell}$ rely on the ZTRAX data set aggregated to the zip code-by-year level by taking the average. Buyer Liquid Assets (y) is measured as the average ratio of total adjusted gross income to total tax returns across zip codes and years, based on the IRS SOI data set. Probability of Delay (q^d) is measured as one minus the share of home sales not settled on time across months, based on the NAR RCI data set. Mortgaged Offer Share (m) is measured as the average share of offers that are mortgage-financed, based on the Offer-Level data set aggregated to the MSA level. Maintenance Cost ($\tilde{\delta}$) is measured as the average ratio of Zillow's ZORI rent index to Zillow's Home Value index across zip codes and months. The table reports the annualized $\tilde{\delta}$ that comes from multiplying by 12. Panel B reports the distribution of the composite parameters ω and L implied by Panel A. Details are in Section V.A.

Parameter	Source	Percentiles		
		15th	50th	85th
	Panel A: Directly calibrated			
Probability of failure (q)	HMDA Denial	0.01	0.047	0.133
Offer arrival rate (λ)	Offer-Level Data	0.871	2.232	3.383
Seller financial wealth (w)	SCF Total Wealth	0.338	1.025	3.369
Down payment (ξ)	Zillow, ZTRAX	0	0.046	0.109
Seller LTV ratio (ℓ_s)	Zillow, ZTRAX	0.042	0.248	0.593
Buyer liquid assets (y)	IRS Income	0.227	0.548	1.508
LTV constraint ($\bar{\ell}$)	Zillow, ZTRAX	0.8	0.88	0.913
Probability of delay (q^d)	NAR Index	0.24	0.3	0.36
Mortgaged offer share (m)	Offer-Level Data	0.354	0.639	0.872
Maintenance cost ($\tilde{\delta}$)	Zillow, ZORI	0.035	0.054	0.079
	Panel B: Composite parameters			
Seller net wealth ($\omega = w - \ell_s - \xi$)		0.044	0.731	3.075
Buyer leverage capacity ($L = y/[1-\bar{\ell}]$)		0.498	1.22	2.335
	Panel C: Baseline preferences			
Coefficient of relative risk aversion (γ): 5				
Discount rate, annualized (ρ): 0.04				
Borrower loan cost, share of par (D): 1				

(a) Table VII: parameter distribution.

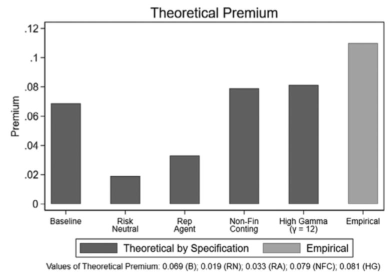

Figure 3. Summary of theoretical mortgage-cash premium. This figure summarizes calculations of the mortgage-cash premium, μ, from Proposition 1. "Baseline" denotes the average value of μ across the empirical distribution of parameters shown in Table VII and the preference parameters shown in that table. "Risk Neutral" denotes the analogous calculation when setting the coefficient of relative risk aversion, γ, equal to zero. "Rep Agent" denotes the value of μ evaluated at the average value of each parameter, not the average of μ across the parameter distribution, and γ again equals its baseline value of 5. "Non-Fin Conting" denotes a similar calculation as in the baseline case after setting the share of mortgaged transactions with a home-sale contingency, h, equal to 7% and the probability of failure due to the home inspection contingency, q^c, equal to 1% (National Association of Realtors 2017, 2018). Equation (38) shows the combined failure rate with contingencies. "High Gamma" denotes the calculation when $\gamma = 12$ and, as in the baseline, μ is averaged across the parameter distribution. The rightmost bar marks 11%, corresponding to the empirical premium. Additional details are in Sections V.A and V.B.

(b) Figure 3: theoretical premium.

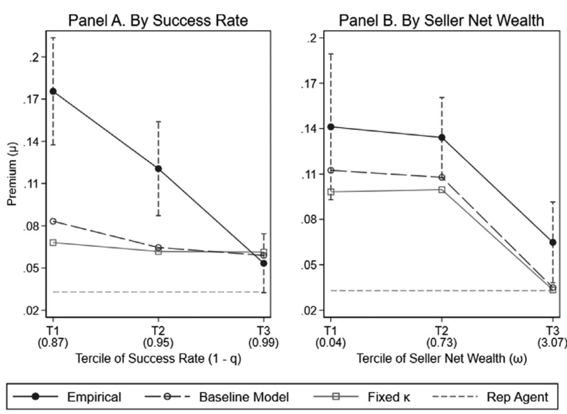

Figure 4. Heterogeneity in the empirical and theoretical mortgage-cash premium. This figure plots the empirical and theoretical premium across the model's parameters. The empirical premium is estimated as in equation (39),

$$\log\left(Price_{i,t}\right) = \hat{\mu}_0 Mortgaged_{i,t} + \sum_{\theta}\sum_{p}\hat{\mu}_{\theta,p}\left(Mortgaged_{i,t}\times Tercile(\hat{\theta},p)_{z,t}\right) + \psi X_{i,t} + \zeta_{z(i),t} + \alpha_i + \epsilon_{i,t},$$

where $\hat{\theta}_{z,t}$ is an empirical measure for model parameter θ at the zip code-by-year level, $Tercile(\hat{\theta},p)_{z,t}$ indicates whether $\hat{\theta}$ lies in the p^{th} tercile of the empirical distribution, and the remaining notes and notation are the same as in Table IV. The theoretical premium is calculated from the projection

$$\mu_g = \mu_0 + \sum_{\theta}\sum_{p}\mu_{\theta,p} Tercile(\theta,p)_g,$$

(a) Figure 4: heterogeneity in empirical and theoretical premiums.

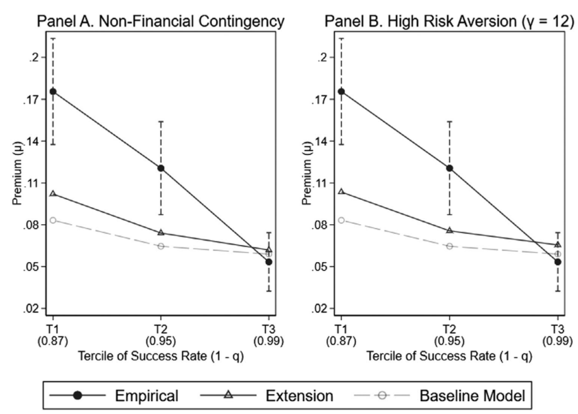

Figure 5. Heterogeneity with nonfinancial contingencies and high risk aversion. This figure is analogous to Figure 4 under two extensions of the baseline model. Panel A incorporates nonfinancial contingencies: Both all-cash and mortgaged offers come with a home inspection contingency, and a share h of mortgaged offers also come with a home-sale contingency. The NAR RCI data set implies $h = 7\%$. The probability of failure due to the home inspection contingency is $q^c = 1\%$, also based on the NAR RCI data set. The probability of failure for a mortgaged offer with a home-sale contingency is shown in equation (38). Panel B uses a higher coefficient of relative risk aversion, $\gamma = 12$. The remaining notes are the same as in Figure 4.

(b) Figure 5: nonfinancial contingencies and high risk aversion.

- The unweighted survey premium is about 10.5%, and Census-weighted premiums are about 10.4% to 10.9% across waves.
- Ambiguous-risk treatments generate higher premiums than known-risk treatments.
- Figure 6 shows that the experimental premium falls with success-rate terciles, resembling the observational pattern.
- When the rate is known, the empirical premium is closer to the model; when ambiguous, the premium is much higher.

Economic message: The survey evidence suggests that belief distortions or ambiguity about failure risk can amplify the premium where transaction risk is high.

Table VIII

Summary of Survey Respondents

This table summarizes the mean of key variables from the experimental survey. Panel A reports the average mortgage-cash premium, elicited through the multiple price list method. The average is calculated (i) without weighting respondents (Unweighted), (ii) weighting by the respondent's representativeness in the 2020 Census according to age, income, home tenure, education, and state of residence (Census Weighted), (iii) after applying Census weights and restricting the sample to respondents with a single switch point between zero and the maximum price in list (Single Switchers), and (iv) after partitioning the single-switch sample according to whether the respondent is told the probability of mortgage transaction failure (Known q) or faces ambiguity (Ambiguous q). All respondents in the first wave face ambiguity. Panel B reports the average model parameters used to calculate the theoretical premium. For respondents facing an ambiguous failure probability, the theoretical premium is calculated using the respondent's subjective probability, elicited at the end of the survey. Similarly, the cost of failure for such respondents equals the subjective value elicited at the end of the survey, and otherwise equals 6%. The seller's financial wealth-to-housing is imputed using the respondent's income and a projection of wealth onto log income estimated in the 2016 SCF data set. Other model parameters are as in Table I. Panel (c) reports the average of demographic variables for the unweighted sample. Section VII and Internet Appendix Section IV contain details.

	Pooled	Average by Wave		
		First	Second	Third
	Average (1)	(2)	(3)	(4)
Panel A: Mortgage-cash premium (μ)				
Unweighted	0.105	0.106	0.104	0.107
Census weighted	0.106	0.109	0.106	0.104
Single switchers	0.096	0.109	0.092	0.087
Ambiguous q	0.103	0.109	0.096	0.096
Known q	0.084	.	0.087	0.081
Panel B: Model parameters				
Probability of failure (q)				
Subjective	0.13	0.135	0.121	0.13
Known	0.07	.	0.07	0.07
Seller cost of failure (κ)	0.051	0.036	0.056	0.06
Seller wealth (w)	2.610	2.513	2.636	2.675
Seller LTV ratio (ℓ_s)	0.3	0.3	0.3	0.3
Seller Down payment (ξ)	0.15	0.15	0.15	0.15
Panel C: Additional variables				
$Household\ Income_k$	\$102,078	\$101,887	\$101,259	\$103,032
Age_k	42.33	40.37	40.99	45.24
$College\ Educated_k$	0.702	0.723	0.703	0.684

Number of respondents in first wave: 1,019
Number of respondents in second wave: 1,202
Number of respondents in third wave: 1,199

(a) Table VIII: survey respondents and experimental premiums.

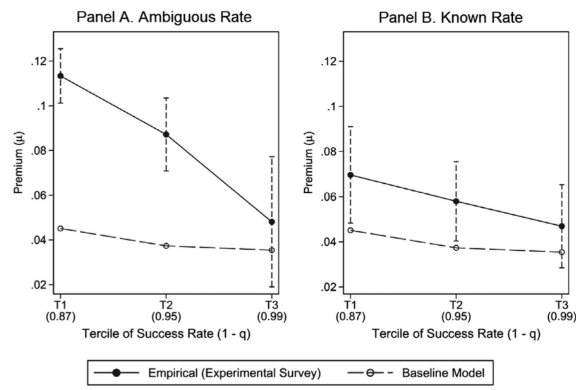

Figure 6. Heterogeneity in the experimental mortgage-cash premium. This figure plots the empirical and theoretical premium using data from the experimental survey, and it is analogous to Figure 4. The theoretical premium is calculated similarly to Figure 4. The empirical premium is estimated as in equation (40),

$$Premium_k = \sum_p \mu_{q,p} Tercile(\hat{q}, p)_k + \sum_p \Delta\mu^A_{q,p} \left(Tercile(\hat{q}, p)_k \times Ambiguity_k\right) + \cdots$$
$$\cdots + \sum_\theta \sum_p \mu_{\theta,p} Tercile(\hat{\theta}, p)_k + \epsilon_k,$$

where k indexes survey respondent, $Premium_k$ denotes the respondent's mortgage-cash premium elicited from the multiple price list, $Ambiguity_k$ indicates whether the respondent is randomized into an experiment in which we do not provide the distribution of q, $Tercile(\hat{q}, p)_k$ indicates whether the respondent's probability of failure, $\hat{q}$ lies in the p^{th} tercile of the empirical distribution used in Figure 4, and we measure $\hat{q}$ using the value of q provided to the respondent or, for those facing ambiguity, the respondent's prior probability, which we elicit at the end of the survey. The vector of indicators $Tercile(\hat{\theta}, p)_k$ has a similar interpretation as in Figure 4. The left panel plots the estimated premium by q when facing ambiguity: $\mu_{q,p} + \Delta\mu^A_{q,p}$. The right panel does so when not facing ambiguity: $\mu_{q,p}$. Respondents are weighted using Census weights to correct for response bias. Standard errors are heteroskedasticity robust. The remaining notes are the same as in Figure 4 and Table VIII. Additional details are in Section VII.C.

(b) Figure 6: experimental heterogeneity.

6 Short summary

- The paper studies why mortgaged buyers pay a large premium relative to all-cash buyers.
- All-cash purchases represent roughly one-third of U.S. home purchases from 1980 to 2017.
- The baseline repeat-sales hedonic estimate is about 11.7%.
- Robustness checks produce premiums between about 8.6% and 16.9%.
- A representative-seller model implies only a 3.3% premium.

- Accounting for heterogeneity in selling conditions raises the theoretical premium to about 6.9%.
- The remaining puzzle is concentrated in high transaction-risk states.
- Survey evidence replicates the empirical pattern and suggests that ambiguity or belief distortions about failure risk can explain the remaining wedge.

7 Conclusion

7.1 Core conclusion

Reher and Valkanov document a large mortgage-cash premium in U.S. housing markets. Mortgaged buyers pay roughly 11% more than all-cash buyers for comparable homes, even after extensive controls and robustness checks.

7.2 Economic intuition

Mortgage financing imposes risk and delay on sellers. Sellers therefore require compensation when accepting mortgaged offers. However, average transaction frictions are too small to justify the observed premium. The key is heterogeneity: sellers in high-risk states rationally require higher compensation, and ambiguity or distorted beliefs about failure risk can further amplify the premium.

7.3 Takeaway

The paper turns a common real estate practice into a pricing puzzle. The difference between mortgage-financed and all-cash offers is not just a small convenience discount. It is a large and heterogeneous wedge shaped by transaction risk, seller financial conditions, and beliefs about failure.